

Responsible use of AI in Evidence Synthesis (RAISE 2026) 2: building and evaluating AI evidence synthesis tools

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Abstract

The evidence synthesis field needs robust, community-driven best practices to facilitate the safe and effective use of automation and AI. Based on the principles of research integrity, the Responsible AI use in Evidence Synthesis (RAISE) project provides recommendations and guidance to ensure the responsible, transparent and ethical use of AI across the ecosystem. It envisions a community of individuals and organisations working in mutually supportive ways to build an environment where reliable evidence synthesis – supported by AI – can thrive.

The RAISE guidance is currently written in three papers. 'RAISE 1', provides tailored recommendations for eight of the distinct roles in the evidence synthesis ecosystem: evidence synthesists, methodologists, AI tool development teams, organisations that produce evidence synthesis, publishers, funders, users, and trainers of evidence synthesis methods.

This paper, RAISE 2, contains guidance on building and evaluating AI evidence synthesis tools. It offers step-by-step guidance on considerations for building different types of AI tools and on how to evaluate them robustly. It contains examples of tools evaluations across different phases of the review process, highlighting how various types of AI have distinct strengths and weaknesses that must be considered in evaluations. It details different performance metrics for consideration, along with references for further reading. Finally, it provides guidance on how to report on and evaluate an AI tool for evidence synthesis.

RAISE 3 provides guidance on assessing, selecting, and using AI tools for evidence synthesis, including ethical, legal, regulatory considerations and the current AI tool landscape.

Highlights

What is already known

- There is excitement about the potential for AI to make the evidence synthesis process more efficient, but also noticeable alarm over the potential for harm
- Guidance is needed to help developers build and evaluate AI tools for evidence synthesis and to inform application of these tools by methodologists and other users

What is new

- This paper provides guidance on developing and evaluating AI tools in evidence synthesis, and how to report both the development and the evaluation transparently
- The three RAISE papers provide recommendations tailored for tool developers, evidence synthesists, methodologists, funders, organizations that produce evidence synthesis papers, and other members of the research community
- RAISE is designed to help the community work collectively to build tools and critically assess tool use to develop an evidence base supporting reliable use of AI in evidence synthesis

Potential impact for RSM readers

- Use of RAISE to guide AI tool development and evaluation should result in more transparent and responsible use of AI in evidence synthesis

Introduction

This is the second of three papers in the ‘RAISE Guidance’ – which aims to support the responsible use of AI in evidence synthesis. The first paper covers *Recommendations for Practice*, providing detailed guidance for each role on the expertise needed to conduct the role effectively in the context of AI. The third paper focuses on *Selecting and utilising AI evidence synthesis tools*. This paper offers guidance on *building and evaluating AI evidence synthesis tools*.

The motivation for this work stems from the numerous requests from the field seeking guidance on when it was ‘safe’ to use AI tools and technologies. This initiated a process of drafting guidance and led to the realisation that the field needed to become more organised before it could produce generalisable guidance about using these tools. Unfortunately, we currently face a situation in which many tools and technologies are being developed and marketed as suitable for use in evidence synthesis, generally without sufficient engagement with the evidence synthesis community. Furthermore, the evidence base supporting the use of these tools is, in many instances, almost non-existent. Where it does exist, it often consists of small ‘N of one’ case examples, making it challenging to draw more general conclusions. Consequently, few tools are used in practice, despite the potential for increased efficiency that automation may offer.

The *Recommendations* paper outlines an ecosystem of individuals fulfilling specific roles, which, if each performs their role, will lead to the creation of tools aligned with the expectations and values of the evidence synthesis community, supported by a robust evidence base of validation studies that can underpin their use. Organisations and trainers will enhance capacity, and the process of

producing evidence syntheses will evolve from a slow, costly, and labour-intensive task into one in which repetitive functions can be automated, allowing human effort to focus on analysis and interpretation.

[How to read this paper](#)

This paper's guidance is split into three sections. Each section is written for specific roles in the ecosystem (as defined in paper 1: recommendations for practice). Each section is also intended to be self-contained, so readers are encouraged to refer directly to the sections that are most relevant to them (though the other sections are likely to be of interest too).

1. Building and validating AI tools

Who is this section for?

- AI tool development teams
- Evidence synthesis methodologists
- Trainers of evidence synthesis methods

Implications of the principles of evidence synthesis for AI tool design

Evidence synthesis is a challenging area for automation, due to its need for higher accuracy and transparency than other AI use cases, as its outputs can influence decisions that affect people's lives. Consequently, comprehensive and rigorous processes and methods have been developed, prioritising accuracy over timeliness and efficiency. When conducted properly, these processes provide a repeatable and transparent foundation for making claims about the status of research knowledge. While decisions in fields that use evidence synthesis methods have always been based on the underlying scientific data, transparency has been a high priority in evidence synthesis approaches since the adoption of systematic review methods for two main reasons. First, systematic reviews aim to reflect the scientific method and be as reproducible as possible. Second, systematic reviews and other forms of evidence synthesis are utilised in regulatory processes where their findings may be questioned by interested parties. For example, having a transparent, relevant, reliable, and auditable process for evidence synthesis is an important aspect in managing challenges and justifying a position.

Although extremely important for evidence synthesis, *transparency* can be a problematic concept in the context of AI technologies for two reasons. First, many AI systems pose challenges for *transparency* (sometimes referred to as 'traceability') because it can be genuinely difficult to explain how a given output was produced. It is thus not surprising that 'Explainable AI' is a growing field of study, due to the relative impenetrability of large neural networks, with even system developers struggling to articulate exactly how they function ¹. Second, commercial interests are frequently cited in response to calls for companies to be more open about how their tools operate. Companies argue that excessive transparency might jeopardise their ability to provide tools at all. When these two factors are considered alongside the evidence synthesis community's natural preference for auditable processes, it becomes clear why the adoption of AI within evidence synthesis has lagged behind other fields ^{2, a}

Therefore, there is a need to reconcile the difficulties that AI systems and tools encounter in providing a transparent and auditable justification for their use with the evidence synthesis community's demands that these tools and processes to embody such attributes. Several approaches can achieve this, and we can learn from practice in other fields. For instance, calls for all tools, data, code and algorithms to be openly and freely available may address some of the desire for openness (though they might not fully resolve the inherent 'black box' issue of current large

^a There is an important distinction between transparency and data security and safeguarding proprietary or personally identifiable information. Data security considerations should be part of building or validating AI tools, and noted in procedures to ensure transparency about the types of data that might not be fully sharable.

language models [LLMs] ¹). Furthermore, in some fields it is quite common to use tools (e.g., drugs) without a complete understanding of exactly how they work, provided that their use is justified by solid evaluations demonstrating that they are both safe and effective for their intended purpose.

A second important principle for evidence synthesis is *comprehensiveness*. Most forms of evidence synthesis (in particular, systematic reviews) aim to provide an accurate summary of the current state of knowledge on a given research question. An evidence synthesis of intervention effectiveness often needs to provide an estimate of the magnitude of an effect, along with its precision. This usually means that evidence syntheses need to find all, or at least as many as possible, and certainly not a biased sample of, the relevant research that has been conducted on a specific topic. The effect of missing studies on evidence synthesis conclusions will depend on the research content of the missing studies relative to the evidence synthesis question, as some studies are more impactful to the development of conclusions (e.g., provide data on primary endpoints rather than secondary or indirect endpoints, have larger sample sizes, have higher study quality, etc.). If an evidence synthesis is later found to have missed relevant research, this can undermine confidence in its findings and may lead to incorrect decisions (e.g., about treatments) being made. Although few studies have been conducted to formally evaluate the impact of missing research studies, comprehensive searches are always preferred. Moreover, such a review might face legal challenges if a company is dissatisfied with a decision about the licensing of its products.

The critical importance of comprehensiveness for evidence syntheses cannot be overlooked also when designing the AI tools. For instance, when developing an application that claims to search for relevant studies, the AI tool development team must ensure that the source (e.g. dataset or repository) used for the search genuinely includes all relevant research for the types of reviews for which their product is marketed. This is challenging, given that many possibly relevant publications are behind publisher ‘paywalls’. However, the existence and impact of ‘Full Text On The Net’ (FUTON) bias has been known about for some time, and the new AI systems based only on open-access papers risk creating infrastructure that generates biased results by design.^{3,4} This important limitation should be clearly communicated to users when search results are generated.

Additionally, searches should be designed to *maximise recall*. Searches that return, for example, the ‘top n’ results – especially when ranked in questionable ways (see *publication bias*) – where ‘n’ is a rather small number, will not be suitable for use in evidence synthesis projects. Historically, evidence synthesists have prioritised recall over precision, and while it is hoped that AI can enhance precision, this should not come at the expense of reducing recall.

AI tool development teams should also be mindful of the risk of *publication bias*.⁵ Publication bias is the failure to publish studies based on the strength or direction of their findings. This results in the tendency for studies with ‘exciting’ or novel results to be more likely to be published than those with ‘null’ findings (it is closely related to various other biases, such as English-language and reporting biases – both of which are related to the systematic preferential publication of positive research findings). Such studies are also more frequently cited.⁶ This means that, in addition to ensuring that the databases searched do not yield biased results, the methods of searching and rating relevance should not inadvertently exacerbate these biases. For instance, using citation metrics to assess the ‘importance’ of search results effectively amplifies publication bias, and must be avoided.

Finally, *accountability* is a crucial principle for the evidence synthesis community. Evidence synthesists are expected to be responsible for their outputs, ensuring that they accurately reflect current knowledge and avoid the biases mentioned earlier. This implies that their AI tools must also be accountable, prompting AI tool development teams to consider the following at each phase of the AI system development:

1. Scoping: ensure constructive consultation from diverse stakeholders, use responsible innovation frameworks and have empowered ethics advisory boards.
2. Data collection: ensure inclusive, transparent, consent-based data collection with privacy and security safeguards, and utilise high-quality, clean data and models.
3. Model training: adhere to rigorous protocols, note potential biases, evaluate security and robustness, ensure algorithm fairness, and confirm staff readiness to execute this effectively.
4. Model testing and deployment: aim for stakeholder feedback, complete proactive testing evaluations, and report publicly, transparently and honestly.
5. Post-launch: ensure continuous auditing of the algorithm and its outputs (including inspecting code and data, testing functionality and accuracy, surveying users' experiences and evaluating metrics, assumptions and social impacts), facilitate validation evaluations, and guarantee transparent reporting of results.

Classes of AI technology

A diverse range of AI technologies exists for various evidence synthesis tasks. These are used in both stand-alone tools for individual tasks and in systematic review platforms that provide “end-to-end” AI-driven assistance for evidence synthesis. We distinguish between classes of *technology* (for example Generative LLMs) and the *implementation* of a technology in a specific tool. This means that we consider the implementation of individual AI-assisted technologies or features (e.g., the Cochrane RCT Classifier) to be a ‘tool’, rather than an entire multi-function platform for facilitating evidence synthesis (e.g., Covidence). What constitutes an individual ‘tool’ varies widely, ranging from an entirely new machine learning model or a new deduplication algorithm, to prompts for using an LLM for a specific task. It is helpful to understand how the different classes of technology operate, as the strengths and limitations of AI tools for evidence synthesis are determined by these inherent capabilities.

Broadly speaking, AI technologies can be considered to fall into five main groups: **rule-based algorithms** (which use fixed rules to make decisions), **unsupervised classifiers** (no labelled training data), **supervised classifiers** (built on labelled training data), deep learning models (neural networks that learn patterns from data), **generative LLMs** (that generate text or responses), and **agentic AI** (end-to-end processes which perform multiple review steps).

Rule-based algorithms are reliable and often transparent. However, they are also “fragile” in situations which lie outside the rules. It is expensive and often not possible to consider every rule that ensures these tools work well in all contexts. For example, most deduplication tools are developed based on sets of rules about the similarity of publication fields. Although many are highly accurate, performance may drop when applied to different publication types (e.g. books versus journal articles) or research fields (where publishing practices may differ). Furthermore, changes to how references are formatted could occur at any time, which pre-determined rules will likely not

accommodate. (Another example of a rule-based algorithm is a Boolean search of the kind often used in systematic reviews.)

Unsupervised classification is used to identify patterns in data without examples or instructions. Such approaches are typically inexpensive (as no training data is required), but the user has no control over which patterns will be presented. An example of this type of machine learning is topic modelling, where passages of text (e.g., titles and abstracts from research reports) can be used to identify recurring themes and topics within the analysed literature.⁷

Supervised classification is the process of training an algorithm on labelled data to make predictions on new, unlabelled data. Supervised AI can be employed, for example, to: distinguish between records that may (or may not) be relevant for a review; identify the type of study being reported; classify population or intervention characteristics; and determine study design traits (for instance, whether participants were aware of their group assignments to assess risk of bias). A notable characteristic of supervised machine learning is that substantial amounts of labelled data are typically required to build the model (e.g., Box 2), which can be hard to come by for some tasks (such as data extraction). They can be expensive to make, often requiring extensive “gold standard” training data curated by human experts. However, these tools can be considered transparent if the training data, model parameters, and adequate documentation are provided.

A specific form of supervised machine learning is ‘priority’ or ‘re-ranking’ algorithms when screening records. Here, relatively few records can be employed as the aim is to have the algorithm rank the remaining records, rather than classify them accurately into two categories. (For such tasks, there is also the possibility of using semi-supervised or, more rarely, reinforcement learning approaches e.g., using word embeddings to cluster records before applying a screening classifier, or optimising study selection based on reviewer feedback.)

Generative Large Language Models (LLMs) are pre-trained on extremely large datasets to generate new content. Although the latest proprietary models come at a financial cost, they do not require large amounts of task-specific training data to deliver useful results. This makes them potentially useful for a wide range of synthesis tasks, including screening, extraction, summarisation, and report writing, with adequate supervision and evaluation. However, this technology comes with several important limitations. First and foremost, there is a risk of ‘hallucination’: the generative AI tool may present incorrect facts or citations. While tools can be powerful for classification tasks, their outputs can also appear confident, yet be incorrect. Based on learnings from their vast training data, these tools also have a “frequency bias” towards dominant pre-existing facts, narratives, and perspectives. This means that while they can be useful for improving reports, using them to generate text can reinforce biases. Compared with other technology classes, transparency is lacking and it is usually difficult to say why a given output was generated. There is also a risk of unintentional plagiarism, where the tool may repeat sections of text from within its training dataset or mirror the structure of given papers. Several recent reviews provide comprehensive overviews of available tools, focusing on specific tasks and the capabilities of generative LLMs.^{8–12}

Agentic AI technologies are behind end-to-end platforms offering to complete a literature review in very short timescales, often from a simple user query. Under the hood, these systems use several generative LLM “agents” to perform sequential tasks autonomously and return outputs to the user. This may include literature searching, screening, data extraction, and summarisation. However, the

limitations which apply to generative LLMs are magnified in this context. Transparency is usually poor, and the complexity of the workflows involved can make it extremely difficult to know exactly how a task was executed or how decisions were made. While such systems may be useful for exploratory work during the planning stage of a review, any reports generated from agentic AI platforms should not be considered rigorous evidence synthesis outputs and must not be used as the basis for clinical, policy, or regulatory decisions.

Building an AI tool

An AI tool is an implementation of one or more of the above classes of technology to perform one or more evidence synthesis tasks. AI tools are developed by individuals in the evidence synthesis ecosystem who fulfil the role of AI tool developer, usually comprising software development teams and computer scientists. Nonetheless, evidence synthesists sometimes build AI tools; for instance, when creating custom machine learning models or writing prompts for a language model (see Box 1). While the users of such tools are typically evidence synthesists, the guidance outlined here applies during the development process.

Box 1: Writing prompts for a language model is building an AI tool

An increasingly common process is designing prompts for a language model to perform a particular task. For example, LLMs are frequently being tested now to assist with screening, data extraction and risk of bias assessment. Sometimes these prompts are automatically generated, and sometimes they are designed manually. In both situations, this process is creating weights, which are then used in a (very large) statistical model. So, when you are designing a set of prompts (for example, to carry out screening or risk of bias assessment), you are designing a machine learning model (manually) for your particular task. This means that the issues raised in this section apply to the apparently ‘simple’ task of building and testing prompts, more advanced processes of creating an agentic workflow, as well as the more conventional process of building a supervised machine learning model from scratch.

Phases of building an AI tool

While many different tools may be described as utilising ‘AI’, creating one typically involves two main phases: 1) the phase when the tool is built, trained, or created (the ‘build’ phase); and 2) the phase when it is tested to determine if it is performing as intended (the ‘validation’ phase). Validating tools to support their use in evidence synthesis must address several dimensions, including reliability, reproducibility (especially for generative systems), and algorithmic bias.¹³ Generalisability or applicability, validation and performance of the AI tool for a particular use case may require additional steps. After deployment, AI tools may enter an ongoing development and monitoring phase, with regular updates in line with new technologies, data structures, and user needs.

The build phase

Building an AI tool can take many forms, including manually creating rules and procedures to follow, writing code to build a machine learning model, or designing prompts for use in an LLM (Box 1). Data is often required when building the tool – either directly in a statistical model – or to assist in its development in other ways (such as designing a deduplication procedure).

Data must be high quality – that is, accurately reflecting the relevant categories and contexts of interest – otherwise, any resulting model will reflect this inaccuracy. AI models trained on flawed

data will have issues or biases permanently embedded. Avoiding *algorithmic bias*¹⁴ is a primary concern in machine learning, making careful evaluation of training data and model performance essential. The size of the sample datasets used for training (and validation) is also an important consideration, as is the balance of the dataset (e.g., are there far more irrelevant records than relevant ones?) and diversity (including sufficient examples for categories and contexts) and ensuring that the metrics used are appropriate. While there is no universally applicable rule of thumb to determine how much data is ‘enough’, it is often safe to assume that having more, diverse, and representative data leads to better models and more robust evaluations.

Transparently sharing the training and validation datasets used enables others to interrogate the data and fosters a deeper understanding of an AI system’s limitations. This helps users to determine when to trust AI systems and in which contexts (i.e., determining *appropriate* trust). An AI system may perform exceptionally well on the training data but poorly on unseen data, indicating that it has overfitted to the training set. Using cross-validation techniques, such as k-fold cross-validation,¹⁵ can help ensure that the model generalises well to unseen data, and employing regularisation methods and early stopping during training can help.

Building supervised and unsupervised AI systems

The first issue to consider when training supervised or unsupervised systems is the source of the data used to train (and test) the model. Once data has been sourced (or generated), it is divided into two parts, usually with the larger portion (e.g., 80%) used to build the model, and its performance subsequently validated on the remaining data (e.g., 20%). (Though when developing prompts for LLMs, this ratio can be reversed, with more retained for validation.¹⁶)

Box 2: example of the process used to build the Cochrane RCT Classifier

The Cochrane RCT Classifier is an ensemble of two AI models designed to distinguish between records that describe randomized trials, and those detailing other study designs. The records of interest are the titles and abstracts of study papers, typically reported in PubMed and other databases. There are several features of the training and testing evaluations of this classifier that are noteworthy in the context of robust evaluation.

First, the size and quality of the datasets used are important (though it is not always necessary to have as much data as used in this example). The model was built using a training dataset of 280,000 records that were manually checked by members of Cochrane Crowd and categorized as ‘may describe a randomized trial’ or ‘definitely not a randomized trial’. The accuracy of the Crowd was confirmed, exceeding 99% sensitivity and specificity. The model was calibrated against a well-known testing dataset (‘clinical Hedges’) which has been widely used for testing and checking search filters. Finally, the model was validated against 92,000 RCTs that were included in Cochrane Systematic Reviews.

The model was developed using three large, high quality, non-overlapping datasets. The records used to build the model were not used during the testing and validation of its performance.

Second, the team that built the model involved and engaged with key interest holders to ensure the required standards and expectations for responsible use were met by working closely with Cochrane Information Specialists and the Information Retrieval Methods Group. These interest holders specified – independently from the system developers– the level of accuracy that the classifier needed to achieve before use in Cochrane reviews (99% recall). This is because it is unacceptable to risk ‘losing’ relevant studies from a review due to an inaccurate AI model.

Therefore, the classifier was built and implemented very cautiously, ensuring that any losses were kept to an absolute minimum. While this necessitates some manual work, users considered this a better trade-off than risking the elimination of relevant research from reviews. They also reviewed documentation about the system to help ensure it was used reliably and validly.

Third, because the model was designed to play a critical role in the systematic review process, where studies might be overlooked if mis-classified, the user community demanded a very high level of recall. This resulted in a system with relatively low precision (8%). Despite this limitation, the model is still able to play a significant role in reducing manual workloads by eliminating records that are almost certainly not RCTs before human review is necessary.

The model was developed by engaging key interest holders to ensure it met the required standards and expectations for responsible use.

Finally, the testing and training evaluations have been published in a peer-reviewed journal.¹⁷ Therefore, the methods and findings have been critiqued by the wider field through peer review and are now publicly available for all, promoting trust and confidence in the model.

Building prompts for generative AI systems^b

In the generative AI paradigm, a statistical classification model is not built. Instead, a 'generative Large Language Model' (LLM), which has been pretrained to 'understand' text, is used. This model has been pretrained to user input – often expressed in plain language – and generate an appropriate response. This type of AI has been popularised by the 'ChatGPT' application, published by OpenAI.^c The ability to produce text means that generative models can perform classification tasks similar to those used in supervised and unsupervised AI, but can also engage in other activities, such as writing summaries and extracting information (including numeric data) from research study reports. In contrast to most supervised machine learning, generative models are not guaranteed to produce the same results every time they receive the same inputs; in fact, they are designed not to. For creative tasks, this can be beneficial; however, when using this technology in evidence synthesis, it becomes a significant limitation, as reproducibility is a core requirement of systematic review methodology. Therefore, steps should be taken to minimise variation (by setting appropriate parameters and/or using prompt engineering techniques), and to ensure that results are not only correct once, but across multiple 'runs' of the same question (or 'prompts').

The prompts used to elicit answers from the LLM are typically developed and tested before use. This also requires splitting the data into 'training' and 'validation' sets, although in this case, the set used for 'training' (i.e., prompt development) is often much smaller than that needed for supervised AI. When developing prompts for generative LLMs, researchers typically select a small dataset and iteratively design and test their prompts until they achieve the desired results. For instance, when

^b We use the term 'generative', rather than 'language model' to distinguish the way that, in generative form, a language model behaves stochastically and requires different evaluation approaches compared to when vector embeddings from a language model are used in conventional supervised machine learning. Building the LLM itself is out of the scope of this guidance; we focus here on the downstream use of generative LLMs for tasks in evidence synthesis.

^c <https://chat.openai.com/>

adapting prompts originally designed for human studies for application in animal studies, one team began with a prompt that stated, “What were the main outcomes measured?”. However, after iterative development using their training dataset, it evolved to read “A complete list of all specific biological parameters assessed for experimental groups”. This type of iterative development and testing should occur within a specified *training* dataset, and no data from that dataset should be used to evaluate the performance of the LLM and prompts.^d

Much of the potential power of generative AI stems from the sheer scale of data used to train the models, which are trained on millions of documents and webpages. However, since these training data contain the biases and inequities found in society, they essentially learn to replicate these biases in their output. Even though considerable effort has been made to minimise ‘toxic’ output from LLMs, many biases remain and can be uncovered with relative ease. There are also concerns about possible linguistic biases that models, which favour the English language due to the quantity of training data, may create or reinforce over other languages that are less well represented.^e For example, 89.7% of training data for the popular Llama 2 model was in English, and model performance was only tested using tasks in English.¹⁸ Not only does this represent linguistic bias, but this is likely also to result in the priorities, perspectives and values of high-income countries being reflected in the model. Users and developers of systems should take these concerns very seriously and test explicitly for impacts of bias on evidence synthesis tasks.

The validation phase

When validating the model, it is crucial to ensure that the data used closely resembles the data for which the model is intended, both in terms of diversity and the balance between positive and negative cases. Being mindful of the context of use also applies to the metrics used to check that the model will be fit-for-purpose. The F1 score (see *performance metrics*) is often used to report the results of machine learning experiments, but as it essentially treats sensitivity and specificity as equally important, it may not be an appropriate metric for the evidence synthesis context, where very high sensitivity is often preferred over specificity (see Section 3 of this paper for more on evaluation metrics).

When validating generative AI, two additional issues must be considered: consistency and hallucinations. As mentioned above, this type of AI does not necessarily produce the same result each time it is presented with the same input data. The extent to which this impacts a given tool needs to be factored into the validation by running the same inputs multiple times and recording any variation in results. Language models are also known to ‘hallucinate’ at times; that is, they may generate results that are untrue or not based on the input data. Importantly, hallucinations can occur even when the requested information is present in the provided text, not only when

^d A common mistake when developing prompts for a LLM is to develop and test prompts iteratively within a review, and to include the records used for prompt development when reporting performance statistics. Prompt development should be considered equivalent to training a ‘model’ for the given task, and data used for prompt development should not be used for testing how well the prompts work.

^e Though the possibility of reducing linguistic bias is also discussed in the literature

information is missing. Again, this should be verified when validating results, ensuring both that the results are correct *and* that they are based on the input data, rather than being ‘hallucinated’.

Building a tool involves testing its validity and, if it is ready for deployment, documenting when and how it can be used. Any recommendations for use should be supported by clear results from evaluations that demonstrate its reliability. If this cannot be done, the tool should only be deployed for evaluation and testing – and this should be reflected in the user interface and supporting documentation. Table 1 contains a template of issues to consider when designing and writing up a study that builds and evaluates prompts for use in evidence synthesis. Additional issues to address include explainability and interpretability, especially given the lack of transparency provided by some genres of AI. Finally, it is useful to describe the tools know or likely generalisability (e.g., based on the limitations or diversity of the training and test datasets).

Table 1: issues and considerations for building and evaluating the use of LLMs in evidence synthesis ¹⁹

Topic	Issues
Research question	Given the variability in responses from generative LLMs, it is important to investigate variability as well as accuracy of these responses.
Dataset(s)	Data must reflect the variability in the domain of interest. Data records used in evaluation must not have been used at all in prompt development.
Prompt development	Researchers often iteratively design and test prompts and sometimes use chains of prompts within the same ‘conversation’ with the LLM. All of this can affect performance and replicability, so needs to be described in detail.
How do you assess accuracy?	It can be difficult to assess accuracy of LLM output, as the same prompt can yield different outputs – but this output may, or may not, differ in meaning. There are some automated approaches (e.g., BLEU and ROUGE scores), but these are functions of % word overlap and may lack sensitivity to important errors. High quality human assessment is helpful, though hard to obtain in high volume. Accuracy can also be assessed using different metrics – e.g., % accuracy or using different ordinal scales. The choice of reference standard is important as this is often used to calculate accuracy measures. A reference standard that contains errors will impact the accuracy scores.
Response stability	LLMs can give different output to the same prompt on repeated tests. Evaluations should assess the consequences of this behaviour.
How are data analysed?	Many statistical and qualitative approaches might be used. Quantitative metrics are often used (see Section 3), but qualitative assessment of accuracy often also plays a role.
How do you interpret the results?	Evidence syntheses conventionally require a high degree of accuracy, as their results often affect decisions that affect people’s lives. The critical question to ask when considering the use of AI is whether its use might

	increase the risk that the review will generate wrong or misleading conclusions.
How and when should the tool be used?	If the results of the evaluation are sufficiently positive, it may be possible to recommend that the tool is used in evidence synthesis. If it is, then the way(s) that it can be used, and its limitations should be documented clearly. There may be situations where, for all the effort that has gone into building and validating a tool, it is not quite reliable enough to be deployed. A good example of a hard – but correct – decision of this kind has been shared by Covidence.^f
Have limitations or generalisability been assessed?	If the tools performance was better in some contexts or areas these strengths and limitations should be noted. Limitations in the datasets, domains, diversity of data, sources, geography, language, publication type, topics, methods, etc. that could contribute to generalisability should be noted.
Reporting	It may be helpful to consider documentation necessary for transparent reporting on AI tool development and validation as the work is planned. Detailed reporting from specifics of the training and test data sets to each topic above will support informed use, replicability, and ability to assess performance across different domains and applications.

^f <https://www.covidence.org/blog/ai-screening-automation-systematic-reviews/>

2. Conducting evaluations to build a cumulative evidence base

Who is this section for?

- Evidence synthesis methodologists
- Evidence synthesis producing organisations
- Trainers of evidence synthesis methods

The importance of evaluations, and who should carry them out

In our model of a role-based ecosystem, evaluations may be conducted by the developer of the AI system or by evidence synthesis methodologists developing best-practice guidance. We deliberately make this separation of responsibilities between roles, although we recognise that they are sometimes fulfilled by the same person (or team), as evaluations are often carried out during a review. However, the task of conducting a review is quite distinct from that of performing an evaluation, requiring different skills and having distinct objectives. We consider it critical for teams to ensure they possess the necessary skills and resources to conduct evaluations reliably, as evaluating performance will always require some additional effort. This also implies that it may not be possible to benefit from the efficiencies an AI tool might offer while simultaneously evaluating it.

Ideally, a tool will undergo evaluations conducted by the developer team prior to its launch, and independent evaluations carried out by evidence synthesis methodologists shortly afterwards (or during ‘beta’ testing). Although evaluations are not always technically complex, they do require careful planning and consideration to select an appropriate study design for the type of AI tool being evaluated. Evaluations, especially those carried out within the context of a single review, should use widely used and accepted methodological ‘templates’ so that their results can be combined. This will enable the field to establish a generalisable evidence base to support the broader application of AI where appropriate. Templates are currently in development to support ‘studies within a review’, or ‘SWARs’.²⁰ SWARs necessitate their own registered protocol,²¹ and a valuable resource for prior SWARs is the ‘SWAR Store’ on the Queen’s University website.⁸ There is a growing interest in conducting SWARs, with one of the most significant areas of expansion being the evaluation of AI tools. That said, at present there is often little in the way of an evidence base to support generalisable conclusions about the applicability and performance of a new tool across different topics and domains. In this situation, evidence synthesists and methodologists may need to validate a tool within the review that it will be employed in. It is possible to do this, and living systematic reviews are a use case where this approach may be particularly useful, though methods are still under development.¹⁶

The use of robust evaluation frameworks is critical for validating the practical utility and reliability of AI tools in evidence synthesis. Randomised controlled trials (RCTs) that directly compare AI-assisted

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<https://www.qub.ac.uk/sites/TheNorthernIrelandNetworkforTrialsMethodologyResearch/SWATSWARInformation/Repositories/SWARStore/>

evidence synthesis with fully manual approaches—or different AI systems against one another—are essential for generating high-quality evidence. Such comparative evaluations not only highlight performance differences and trade-offs (e.g., accuracy versus efficiency) but also help identify contexts in which AI applications are safe, effective, and ethically sound. Moreover, embedding SWARs provides an opportunity to systematically evaluate AI tools under realistic conditions, simultaneously advancing methodological rigour and contributing to a cumulative evidence base. Adopting standardised methodologies and reporting protocols for these evaluations ensures consistency, reproducibility, and transparency—attributes highly valued by the evidence synthesis community.

Considerations for evaluations of AI performance on different tasks

AI tools are available for nearly every phase of the review process, making it vital to tailor the evaluation method to the specific task at hand. In addition to the metrics outlined below it is essential to consider whether using a tool genuinely enhances efficiency in real-world settings. Evaluations must take this into account, and collecting ‘time-on-task’ data, which allows for the comparison of manual and (semi-) automated approaches in terms of human effort, can be very beneficial. This method aligns closely with the goals of SWARs and can be incorporated directly into systematic reviews to provide realistic, context-sensitive insights into AI performance. Covered below are the tasks for which tools are being actively developed and evaluated. We have not yet seen very much activity for later review stages – such as analysis or assessing the certainty of evidence – but no doubt these will be covered in time. The same overarching concerns apply throughout, though, that a good evaluation usually depends on a good “gold standard” dataset which serves as a comparison to measure the AI output against.

The Search

Increasing numbers of tools conduct ‘semantic’ searches for studies from their own datasets of millions of research studies. The first consideration when evaluating such a tool is the source of its data: does it contain all the records likely to be relevant in each review? (Please see also *Implications of the principles of evidence synthesis for AI tool design*, above for further discussion on how search systems can introduce bias.) Many bibliographic databases utilise large open-access catalogues of journal literature, allowing the scope of the research they include to be assessed by comparing them to the journals covered by databases that would otherwise be searched using conventional methods. Although databases typically describe their research coverage areas (e.g., PubMed focuses on biomedical and life sciences research), they tend to be English-language biased, and identification of non-English language publications may require additional steps. In addition, grey literature is typically less well-covered and harder to evaluate.

If an AI tool contains all the research likely to be relevant, the next consideration is whether the right research can be identified using the AI search features. The methods commonly used to evaluate search filters can also be applied to assess how effectively an AI search tool finds research.²² A set of ‘gold-standard’ records or a reference (or ‘benchmark’) set (i.e., records that should definitely be identified, and ideally representative of all relevant records), is needed. The tool can then be evaluated based on the number of gold standard records it retrieves (sensitivity) against the number of irrelevant records it also identifies (specificity).^{22,23} More specific details regarding approaches to building reference sets (such as the use of a relative recall approach) can be found in Lefebvre 2017

and Sampson 2006^{22,24}. For guidance on evaluating AI search tools, refer to the Evaluation Instrument constructed by the Canadian Drug Agency (CDA-AMC)²⁵. This Instrument is the result of a systematic examination of 17 tools and encompasses a range of search-related tasks, including topic clarification, search strategy development, search execution, and strategy documentation.

Search tools extend beyond those that carry out searches to tools that assist in the ‘translation’ of searches between databases (e.g., Polyglot²⁶) and those that help with citation checking and ‘snowballing’ around known studies.²⁷ Again, when evaluating the utility of these tools, a suitable ‘gold standard’ set of records should be used. In the case of search translators, the key question is not only whether the search is successfully translated, but whether it maximises its use of the features available in the target database. For example, the translation of advanced operators (e.g., proximity and NEAR) may be extremely powerful but are also tied quite specifically to search engines. Likewise, while there are a few ‘citations-chasing’ apps available now, they may vary in their ability to recall all citation connections (for example, some commercial databases appear to have more comprehensive citation networks than open-access ones). Recent evaluations of novel technologies for searching include: ^{28–33}.

Deduplication

Many papers evaluate algorithms for identifying and removing duplicate citation records. There are numerous open-access datasets containing test cases of duplicate records, enabling head-to-head comparisons. One consideration to bear in mind when evaluating a tool (or critically appraising another evaluation) is whether the open-access dataset used for evaluation was also utilised in the tool’s development. If so, performance is likely to be overestimated. For examples of evaluating deduplication tools, see ^{34,35}. Please note that, as with devising prompts for an LLM, records used to develop and test a deduplication tool should not be used to evaluate its performance. Also, please note that duplicates in an evidence synthesis project rarely occur in pairs and that groups (or clusters) of duplicated records are common, complicating the duplicate discovery and evaluation task.

Screening search results

In cases where the AI tool classifies records as potentially relevant or irrelevant, evaluation must start with a dataset of gold standard records that encompasses the full scope of the use case in question.³⁶ These can be processed through the machine learning tool, aiming to determine whether sensitivity is sufficiently high (possibly reaching 100%, depending on the use case).

A special case of supervised AI is the use of algorithms to prioritise the order in which records are viewed for title and abstract screening. Again, retrospective evaluations are possible, providing evidence about how well the prioritisation might have performed (if it hadn’t been used previously), or how effectively it probably worked in the relevant evidence synthesis. Many papers on this popular machine learning capability have been published, with research dating back to 2006³⁷, and a systematic review of these studies was published in 2015.³⁸ A key unknown now, however, is how to evaluate, during the screening, when a given level of recall has been achieved, and research in this area is ongoing.³⁹ For example, no one has formally evaluated the impact of excluding studies due to a decision to halt screening at a predicted proportion of items found (e.g., 98% relevant references) on a review’s conclusions. Similarly, no one has evaluated the relative utility or impact of studies prioritised early and included, compared to those prioritised later and included.

Data extraction and risk of bias / critical appraisal assessment

Evaluation methods for data extraction and risk of bias assessment often mirror those mentioned earlier, particularly concerning the availability of ‘gold standard’ data for assessing the performance of the LLM. When evaluating data extraction automation and assessing the risk of bias, several additional considerations should be taken into account.

First, this task is often done by (at least) two humans working independently because it is difficult to achieve high levels of accuracy, and even with this dual approach, errors are still likely ^{40,41}. It should also be noted that other factors, in particular incomplete reporting in the extracted documents, may lead to inaccuracies in the results.⁴² In some cases, effective risk of bias assessment of a given study may require additional content (e.g., on the study population or geographical location) that is not included in the study. Thus, ensuring that evaluation data are as accurate as possible can be challenging and time-consuming; it is unlikely that any review evaluating automated data extraction will also benefit from its efficiencies.

Second, verifying the agreement between AI and humans is more challenging automatically, and this often requires manual intervention. In addition, data extraction evaluation involves defining error type and severity, and determining the appropriate unit of analysis for composite data. Please note that automated metrics, such as BLEU and ROUGE scores, which claim to be able to provide automatic evaluations of extracted text, do not appear to work well in the evidence synthesis context.¹⁹

Third, one concern regarding automation is that it can lead to less accurate work by human evidence synthesists. ‘Non-inferiority’ studies, which aim to check that AI-assisted data extraction and risk of bias assessment are at least as good as humans alone, seek to address this question.⁴³

Fourth, there are concerns that open-access reviews should not be used to validate LLM-based tools, as the reviews could, in principle, have been included in the training data, leading to data contamination. However, there are examples of randomised and non-randomised trials conducted that can serve as templates for future evaluations.^{43–50}

Fifth, similar to the variation in entity data extraction, there is a large degree of variation in the complexity of risk of bias questions or domains. Therefore, there is an expectation that AI tools will perform better on some risk of bias questions than on others. For example, whether human subjects or experimental animals were randomised to a treatment group is an important consideration in many risk of bias tools. Randomisation can be assessed by searching for relatively few words in a publication, and it is a simpler concept than assessing confounders, which are complex and likely to differ considerably across studies and journals in how they are described.

Finally, whilst there is currently great interest in the potential of LLMs to assist in data extraction and risk of bias assessment, generative AI is known to be prone to ‘hallucinate’ – i.e., generate false content – and it is possible that, if the required data is not present in the document presented to the LLM, it will manufacture erroneous data. It is difficult to evaluate the extent to which a given tool may be prone to this kind of error (though see *Specific considerations related to evaluating Generative AI* below), and it may be sensible to ensure that the relevant information is present before prompting an LLM to extract it.⁴²

Text generation

When it comes to evaluating the accuracy of text, automated methods tend to fall short because of the many ways a given fact can be expressed. This means that comparing the ‘similarity’ of text using simple similarity metrics is usually insufficient. Metrics such as ‘BLEU’ and ‘ROUGE’ have been suggested (see below), but their performance has been found to be poor. Using LLMs as ‘judges’ has been proposed as one solution, in which one LLM assesses the performance of another. As the alternative is for humans to check every result, LLMs as judges offers a more scalable way to evaluate LLM-generated text, though it is not (yet) an approach that has seen widespread use in evidence synthesis.

‘Agentic’ tools which combine multiple review stages

Some tools undertake multiple review stages in ‘agentic’ workflows. For example, following a user query, they will search for research, screen search results, extract data, and even attempt to synthesise results. There are two main ways of evaluating such systems. First, each stage can be evaluated in isolation. For example, using the approaches above to evaluate the extent to which the search retrieved the right records and the automated screening identified those eligible for the review. Second, it would be possible to evaluate the final stage of the process independently. Were the right studies identified, and do the findings of the LLM-generated report agree with a human-generated synthesis? One critical point to bear in mind is that this evaluation will yield invalid results if the LLM’s training data included the review(s) used in the evaluation.

3. Performance metric considerations for AI tool evaluations

Who is this section for?

- Evidence synthesis methodologists
- AI tool development teams
- Trainers of evidence synthesis methods

Choosing appropriate performance metrics

Researchers and methodologists should utilise validated measures to assess the performance of AI tools. A key aspect of this involves using appropriate benchmarks. In evidence synthesis, benchmarks for tool performance may vary by context of use. This context can refer to the type of review, domain area, or the intended mode of tool implementation. No tool will achieve both perfect accuracy and high efficiency in all (or even most) scenarios. However, this does not imply that the tool is unusable. The goal is to identify methods for safe implementation that significantly increase efficiency while maintaining the overall accuracy and integrity of the evidence synthesis. The essential question then becomes: what defines ‘good enough’ in terms of tool performance? While it is the impact of bias on the conclusion of an evidence synthesis effort that is the most important, it is often easier to assess performance of AI methods for individual steps such as literature search or data extraction.

When completing evaluations, we recommend that both the evaluation metrics and the generated datasets be made available, so they collectively contribute to the cumulative evidence base on the use of AI in evidence synthesis (see Section 2). This will support evidence synthesis methodologists in moving towards standard methods and protocols, and the use of standardised performance metrics (or core outcome sets), which are fundamental to the reliable implementation of AI in evidence synthesis.

Specific considerations related to evaluating Generative AI

The advent of generative AI has dramatically expanded the types of tasks and the number of reviews where AI could have an impact. This makes evaluation more challenging and may also hinder our ability to generalise or validate externally. Over the coming years, the way we evaluate generative AI will likely need to evolve in response to its growing capabilities. We may witness a shift from performance-oriented evaluation to capability-oriented evaluation.⁵¹

Along with the expanding capabilities and task space, there are three key considerations related to generative AI: stability, robustness, and the potential for hallucinations.

Stability: A key question to ask is, “even if the system gets the correct answer sometimes, how can we be sure that when we use it, we won’t be affected by some random variability?” Evaluating stability can help us to avoid systems that either lack evidence of their stability or are clearly too unstable for use in evidence synthesis. There are few established metrics for evaluating stability, however, estimating the frequency and seriousness of deviation, may serve as useful starting points. For instance, in a classification use case where the LLM determines whether a study belongs to a

specific category, a practical evaluation approach might involve running the same data through the LLM under identical conditions and recording how often the predicted categories change. (Care must be taken when conducting such evaluations that they are not misled by LLMs remembering previous requests and simply sending back the same response without it having been regenerated (known as ‘caching’).

Robustness: Even within the context for which the system was designed, there will be ‘edge cases’ that may lead to unexpected responses from the system. Similarly, a system’s ability to handle unexpected inputs in a predictable manner should be among the aspects tested and validated during evaluation. At times, human judgment and interpretation, along with metrics, may be necessary to bring essential nuance to these assessments.

Assessing the presence and impact of hallucinations: The presence (or absence) of hallucinations can be evaluated, for example, by directly asking about content that is not present in the data provided, or posing nonsensical questions to see if the system accurately states that the information is not available, or whether it creates fictitious content instead.⁵² When the system is expected to cite its sources, verifying that the cited sources exist and contain the information claimed is an essential check to perform in this context. It should be noted that these approaches only test specific types of hallucination scenarios, and LLMs may still hallucinate in other situations where the requested information is present in the input.

Move beyond absolute true/false classification: Evaluating the performance of LLMs presents challenges inconsistent definitions and applications of evaluation metrics across studies. For example, researchers may label a partially extracted field in a data extraction form as a false negative in one context and a false positive in another, leading to ambiguity and reduced comparability of results. This lack of standardisation complicates efforts to assess model reliability and hinders progress toward robust benchmarking. One suggestion has been to move beyond correct/incorrect assessments and instead conceptualise the differences as *concordant* or *discordant*⁵³, where concordance implies factual alignment between extracted data items, despite differences in formatting or structure.

Assessing the accuracy of generated text: In a text summarisation task, where the requirement is to outline the content of an intervention, quantifying stability is more challenging. The extracted text may vary from one ‘run’ to the next, but possibly not in significant and meaningful ways. While there are some automatic tests (e.g., *BLEU* and *ROUGE* scores – see below), these do not necessarily capture important actual differences. The best approach currently may be to run the same data through the system and manually assess whether differences exist and if they are meaningful, or merely variations in phrasing. Additionally, using LLMs as ‘judges’ to evaluate semantic similarity between different outputs may provide a more scalable approach to stability assessment.

Common performance metrics

A comprehensive evaluation framework for AI tools in evidence synthesis requires considering multiple dimensions beyond simple accuracy metrics. To better structure these evaluation approaches, we propose the following taxonomy of metrics (Figure 1).

Taxonomy of common performance metrics for AI tools for evidence synthesis				
1. Accuracy metrics	2. Reliability metrics	3. Efficiency metrics	4. Usability metrics	5. Risk assessment
1.1 Classification performance (confusion matrix, sensitivity, specificity, accuracy, F1 scores) 1.2 Content generation quality (text and semantic similarity, completeness, correctness)	2.1 Variability (variation from the mean) 2.2 Consistency (inter-rater reliability, response stability) 2.3 Robustness (input robustness)	3.1 Resource utilisation (cost-effectiveness, workload reduction, time efficiency) 3.2 Workload impact (time-on-task, human effort required for verification)	4.1 User experience (user satisfaction and usability) 4.2 User integration (implementation barriers, interoperability measures)	5.1 Error impact (error severity scoring, missed critical information, changes in synthesis result) 5.2 Safety and ethics (transparency, bias, environmental impact)

The taxonomy includes some of the more commonly used evaluation metrics and statistical tests. It is not intended to be prescriptive but to help evidence synthesis methodologists to select appropriate evaluation methods systematically and consistently, based on their specific context, the task the AI system is being applied to, the type of AI being assessed (e.g., unsupervised, supervised or generative), and the team’s particular evaluation goals. Please see Appendix 1 (below) for full definitions, examples and references for each category.

4. Reporting the building and evaluation of an AI tool

Who is this section for?

- AI tool development teams
- Evidence synthesists
- Evidence synthesis methodologists
- Trainers of evidence synthesis methods
- Organisations that produce evidence synthesis

Reporting guidelines and guidance

Reporting transparently on how an AI tool was developed and validated is crucial for informing adoption decisions and ensuring that methods for evidence synthesis advance reliably. It also facilitates replicability, further verifying performance across different domains and applications. While there are not yet widely adopted reporting standards for AI tools in evidence synthesis, there are numerous reference points for reporting the development of AI tools for practical applications. For example, Kolbinger and colleagues⁵⁴ identified 26 such guidelines in healthcare, and many papers reporting principles and high-level guidance on the use of AI in different settings.

It is important to note that many types of evaluations are conducted on AI tools, including: (1) performance evaluation that addresses how well AI performs; (2) process evaluation that addresses how well AI works in realistic settings; and (3) human evaluation that address how the use of AI affects the human processes involved in evidence synthesis. We focus here on the reporting of performance and process evaluation but note that human evaluation is also important when considering the adoption of AI tools. Below, we adapt the framework proposed by Kolbinger and colleagues for use in the evidence synthesis context, aligned with recommendations throughout RAISE for different roles, and suggest that authors adapt the tool as needed to suit their specific use case.

Introduction

Existing tools and knowledge (optional). Authors should provide an overview of the current state-of-the-art and how their new study or tool contributes new knowledge or capabilities to the field.

AI tool. Include details on the AI system, including:

- AI tool name and version
- Developer name and country
- How to access it and any user guides and support available.
- How the tool works and how it achieves its intended goal(s). Include what stage of evidence synthesis are affected, whether it should replace an existing task or method or if it is to be used in parallel with current methods. If the latter, state which existing task or method this new approach should complement or be used alongside. Confirm how the AI system is setup so humans have control of its operations and outputs (i.e., how it has implemented human oversight) or state this is not relevant and why.

Objective. Authors should give an overall statement on the objective(s) or research questions(s) the evaluation addresses.

Methods

Study design. Authors should state whether their validation study was prospective or retrospective, and the nature of the comparisons made (if any).

Setting. Authors should state the domain that the tool was developed in (e.g., public or environmental health)

Data sources. Authors should state where their data came from, and its structure (where relevant). Data should adequately reflect the variability of the setting of interest. A specific statement that distinct datasets were used for training, testing and validation stages should be included, if applicable.

Data selection. Unless authors used all existing data, they probably used a sample. What were the inclusion and exclusion criteria used, and how were they applied? How did authors ensure the dataset was adequately sized and sufficiently broad to cover the stated topics and domains of the setting adequately?

Data preprocessing. How were data transformed (if they were), and how were missing data and outliers identified and addressed?

Labelling of input / validation data. How were 'gold standard' data created? How was accuracy of this dataset measured and assured?

Type of AI tool / model. State the type of algorithm and foundation models (e.g., OpenAI's ChatGPT, Anthropic's Claude, etc.) (if used), what processes are involved in integrating and deploying the tool (where relevant).

Tool development. How was the tool (or model or prompts etc.) developed? Prompts that are designed and tested iteratively should be described in detail as it this can affect performance and replicability. What processes were followed to ensure train / test data were not contaminated and to ensure the process was replicable and robust?

Performance accuracy: Report which outcomes are measured, metrics used and define comparators. Report on issues relating accuracy and variability, range or confidence, especially with generative AI which will have variable responses.

Validation methods. State the statistical and / or qualitative approaches to analysis, including performance comparisons and baselines. What methods were used for assisting internal and external validity?

Performance errors. How were errors identified, and what error analysis was conducted?

Interpretation.

Discussion

Strengths and limitations. This relates to the domains and breath of the data used for training, testing and validation, and so how generalisable the AI system is across different types of evidence synthesis. Consider aspects such as scope, sources (e.g., [PubMed only, etc](#)), geography, language, methods, publication types, etc., as well as the data sampling and processing methods. Which types of evidence synthesis might it be best suited for / less useful for / inappropriate to use? Are there limits in terms of its generalisability (e.g., to different domains)? If an evidence synthesist was using this evaluation to decide whether to use an AI system, what interpretation pitfalls do they need to be aware of?

Bias. Explicitly state which biases or equity issues might affect the use of the tool or model. What do users need to be mindful of when using this tool? Are there any strategies for users to help mitigate potential issues due to biases, e.g., were data obtained from published sources only, or open-access sources only.

Practical value of the tool. Does the AI system augment current workflows, and if so, how might findings and conclusions benefit from its use? How should this tool be used (if it should be) in production evidence synthesis workflows? Detail the cost to implementation, integration, and / or use.

Implications for use. How should the results of the validation be interpreted for use? A critical consideration is whether the use of the AI tool has the potential to invalidate the evidence synthesis findings or make the conclusions unreliable, therefore detail the potential risk of this and under what circumstances. Confirm any requirements for how it should be reported or visualised.

Other information

Ethical statement. Provide details on ethics approvals (where relevant) and a statement on the tool provider's ethical stance with regards to LLMs (where appropriate)

Availability of protocol. Authors should state whether a protocol was publicly available prior to the evaluation being conducted and, if so, where it can be found.

Sources of support. Describe sources of financial or non-financial support for the evaluation, and the role of the funders or sponsors in the evaluation.

Declarations of interest. Declare any financial or non-financial interests of the evaluation authors, including whether the authors are independent of the AI tool developers. Include a statement about whether the tool is produced by a commercial organization that can gain financially from use of the tool; gain financially could include selling it, licensing it, making it available through paid subscription, etc.

Availability of data, code and other materials. Report which of the following are publicly available or where they can be found, e.g., on reasonable request: training dataset; testing dataset; prompt development; evaluation dataset; performance analyses data; analytic code; any other materials used in the evaluation.

Replicability. State how the study results can be replicated by 3rd parties. (It is possible that they cannot be, but this statement needs to make this clear.)

Environmental impacts. State any relevant considerations about environmental impacts of the tool i.e., energy, water use, emissions (which is related to computational power or efficiency).

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Appendix 1: full details of the taxonomy of performance metrics

1. Accuracy metrics

General principles for all accuracy metrics

- The performance of any AI tool can only be as good as the reference standard (gold standard) against which it is evaluated
- Dataset size impact all classification metrics - larger, balanced datasets provide more reliable estimates
- High-quality, accurately labelled validation datasets are essential for meaningful accuracy assessment

These principles apply to all accuracy metrics described below.

1.1 Classification performance

Confusion matrix	
Related terms / metrics	NxN matrix, Error matrix, 2x2 table
What does it measure	A <i>confusion matrix</i> can be used to calculate several metrics including accuracy, sensitivity, specificity and precision (all of which are described in more detail below). The matrix is made up of values with four possible designations: True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN).
Considerations for use	• Often used to evaluate binary classification models • Usually used as a basis for further calculations e.g., <i>sensitivity</i> and <i>specificity</i>, Youden's Index ⁵⁵ See metric overviews in Rainio 2024, Walsh et al 2021, and Berrar 2019 ^{55–57}

Sensitivity	
Related terms / metrics	Recall, True positive rate
What does it measure	<i>Sensitivity</i> measures how well a test (or an AI system) can correctly identify the true positives, i.e., the proportion of relevant items in the output/dataset that are correctly identified by the AI system compared with the total number of genuine items. The formula is: $TP/(TP+FN)$
Considerations for use	• <i>Sensitivity</i> is often presented alongside <i>specificity</i>, with greater importance usually placed on higher sensitivity, rather than higher specificity • However, for evaluating performance of AI for data extraction, sensitivity does not reflect the extent of incorrect data extraction such as fabricated data for items for which no data were available (as this would normally be labelled false positive). Potentially useful new metrics include “faithfulness” ⁵⁸ and “confabulations” ⁵⁹. See metric overviews in Rainio 2024 ⁵⁶, Walsh et al 2021 ⁵⁷, and

	Berrar 2019 ⁵⁵ . Also see review of data extraction tool performance by Schmidt et al 2025 ⁶⁰
Example(s) of its use	• Evaluation of RCT screening classifier by Thomas et al 2021 ¹⁷ • Evaluation of a data extraction tool by Walker et al 2022 ⁵⁰

Specificity	
Related terms / metrics	True negative rate
What does it measure	<i>Specificity</i> measures how well a test (or AI system) can correctly identify the true negatives. The formula is: $TN/(TN+FN)$
Considerations for use	• <i>Specificity</i> is often presented alongside <i>sensitivity</i>, with greater importance placed on higher <i>sensitivity</i>, rather than higher <i>specificity</i> See metric overviews in Rainio 2024 ⁵⁶, Walsh et al 2021 ⁵⁷, and Berrar 2019 (ref). Also see review of data extraction tool performance by Schmidt et al 2025 ⁶⁰
Example(s) of its use	• Evaluation of RCT screening classifier by Thomas et al 2021 ¹⁷ • Evaluation of a data extraction tool by Walker et al 2022 ⁵⁰

Precision	
Related terms / metrics	Positive predictive value
What does it measure	The <i>precision</i> metric predicts how often the test/tool/model correctly predicts the class of interest. The formula is: $TP/(TP+FP)$
Considerations for use	• Useful metric where the classes are imbalanced as it shows how well the test/tool/model is at identifying the class of interest • Does not consider False Negatives • Often used instead of <i>specificity</i> • For data extraction, it does not take missed data into consideration and can be high even if the AI missed a lot of information See metric overviews in Rainio 2024 ⁵⁶ and Berrar 2019 ⁵⁵. Also see review of data extraction tool performance by Schmidt et al 2025 ⁶⁰
Example(s) of its use	• Evaluation of RCT screening classifier by Thomas et al 2021 ¹⁷ • Evaluation of a data extraction tool by Walker et al 2022 ⁵⁰

Accuracy (and error rate)	
Related terms / metrics	Percentage agreement

What does it measure	This metric expresses the proportion of all correctly classified instances. It is calculated using the four values or designations from a confusion matrix, True Positive (TP), True Negative (TN), False Positive (FP) and False Negative (FN). The formula is: $(TP + TN)/(TP + TN + FP + FN)$. With the <i>accuracy</i>, you can also estimate the error rate ($1 - \text{accuracy}$).
Considerations for use	• Often used to evaluate binary classification models • Providing an accuracy score alone can be potentially misleading or simply not helpful in determining whether an AI tool/system is effective. This is especially the case where there is an imbalance in the dataset of the class of interest and negative instances. • <i>Sensitivity</i> and <i>specificity</i> often reveal more about a model's performance (see tables above) See metric overview in Rainio 2024 ⁵⁶. Also see review of data extraction tool performance by Schmidt et al 2025 ⁶⁰
Example(s) of its use	• Evaluation of a data extraction tool by Walker et al 2022 ⁵⁰ • Evaluation of LLMs for detection of publication bias from funnel plots or quantitative values

F1 score	
Related terms / metrics	Dice coefficient, harmonic mean of precision and recall, F-score, F-measure
What does it measure	The F1 score is the harmonic mean of sensitivity and precision. An F1 score ranges from 0 to 1, where 1 indicates perfect precision and recall. The formula is: $2 \times (\text{Sensitivity} \times \text{Precision}) / (\text{Sensitivity} + \text{Precision})$
Considerations for use	An F1 score can be a very useful metric. However, it does have two main aspects to be aware of: • It does not take account of the true negatives. This can be an issue where there is a high prevalence of true negatives • The F1 score assumes precision and sensitivity (recall) are of equal importance – this is often not the case in evidence synthesis uses case, where sensitivity is usually paramount. Due to this, the F-beta score is often preferred (see table below) See overviews of classification metrics in Walsh et al 2021 ⁵⁷. Also see review of data extraction tool performance by Schmidt et al 2025 ⁶⁰
Example(s) of its use	• Used as an accuracy metric for an information extraction algorithm in West et al 2024 ⁶¹ • Evaluation of a data extraction tool by Walker et al 2022 ⁵⁰

F-beta score	
Related terms / metrics	F-0.5, F-2, F-3 score (depends on the specified B parameter)
What does it measure	The F-beta score is a variant of the <i>F1 score</i> (detailed above) that is used to measure the performance of a classification model. The F-beta score enables more importance to be assigned to either precision or sensitivity.
Considerations for use	- The F-beta metric can be particularly useful in imbalanced datasets where one class is more prevalent - The chosen beta value (the parameter that determines the weight of recall) is critical and people interpreting the F-beta score need to understand how the implications of the chosen beta value See overview of classification metrics in Berrar 2019 ⁵⁵
Example(s) of its use	F-3 (recall emphasised over precision) in evaluation of LLM based screening by Wang et al. ⁶²

Root-mean-square of the error (RSME)	
What does it measure	This is an accuracy metric for prediction algorithms, which measures error of prediction as the difference between the predicted and annotated outcome value
Considerations for use	See overviews of classification metrics in Berrar 2019 ⁵⁵
Example(s) of its use	Used as an accuracy metric for a prediction algorithm in Chai 2014 ⁶³

Area Under the Receiver Operating Characteristics Curve (AUC-ROC)	
What does it measure	A receiver operating characteristics (ROC) curve illustrates the performance of a classifier model at different threshold values. Specifically, it plots the true positive rate (see <i>Sensitivity</i>) versus the false positive rate ($1 - \text{Specificity}$). The area under this curve (AUC) is indicative of the model performance. A straight diagonal line (with AUC = 0.5) indicates that the model performs no better than random category assignment, while an AUC of 1 would indicate a perfect classifier.
Considerations for use	- Widely used metric to evaluate and compare the performance of binary classifiers, - Can also be used to evaluate multi-class classifiers (MAUC) ⁶⁴ - Value based performance labels (e.g., 0.8 may be considered “good”) for AUC are largely arbitrary and will depend heavily on the task at hand ⁶⁵ - Handles imbalanced data (common in evidence synthesis) well.

	Not always easy to interpret, and while being independent of decision thresholds is often helpful, this can be a weakness when a specific threshold is needed for decision-making See overviews of classification metrics in Walsh et al 2021⁵⁷. Also see review of data extraction tool performance by Schmidt et al 2025⁶⁰
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1.2 Content generation quality

BLEU (BiLingual Evaluation Understudy) score ⁶⁶	
Related terms / metrics	SacreBLEU, GLEU (Google BLEU)
What does it measure	Evaluates text generation quality by comparing overlap between generated and reference texts. Originally designed for machine translation evaluation.
Considerations for use	- Focuses on exact word matches and may miss semantically equivalent but differently worded content - Schmidt et al (11) evaluated <i>BLEU</i> and <i>ROUGE</i> scores for PICO extraction tasks in systematic reviews and found they were poor indicators of accuracy. This limitation applies broadly to word-overlap based metrics when semantic accuracy is more important than exact wording. - Better suited for tasks where exact wording matters more than semantic meaning
Example(s) of its use	Comparison of human-extracted descriptions of PICOs used in randomised trials with those extracted by an AI tool by Schmidt et al (11)

ROUGE (Recall-Oriented Understudy for Gisting Evaluation) score	
What does it measure	Evaluates text generation quality by measuring overlap of word sequences, and word pairs between generated and reference texts. Originally designed for summarisation evaluation.
Considerations for use	- Like <i>BLEU</i>, focuses on surface-level text overlap rather than semantic similarity - May not capture meaning preservation when different words convey the same information - Schmidt et al (11) evaluated <i>BLEU</i> and <i>ROUGE</i> scores for PICO extraction tasks in systematic reviews and found they were poor indicators of accuracy. This limitation applies broadly to word-overlap based metrics when semantic accuracy is more important than exact wording.
Example(s) of its use	- Comparison of human and LLM generated medical summaries of Cochrane reviews in Tang et al.⁶⁷ - Comparison of human-extracted descriptions of PICOs used in randomised trials with those extracted by an AI tool by Schmidt et al (11)

METEOR (Metric for Evaluation of Translation with Explicit ORdering) score ⁶⁸	
What does it measure	Evaluates text generation quality using a combination of exact text matches, stem matches (e.g., “test” vs “testing”) and synonyms.
Considerations for use	Can be used to evaluate text translation or summarisation tasks
Example(s) of its use	Comparison of human and LLM generated medical summaries of Cochrane reviews in Tang et al. ⁶⁷

BERTScore ⁶⁹	
Related terms / metrics	Semantic similarity score
What does it measure	Computes text similarity scores between generated and reference texts based on meaning, rather than exact word matches (<i>unlike BLUE and ROUGE</i>).
Considerations for use	Well-suited to language generation tasks where semantic similarity is more important than exact wording
Example(s) of its use	Evaluation of LLM-based PICO and study design extraction ⁷⁰

Information completeness	
Related terms / metrics	Difference in level of detail, Coverage assessment, ICAT ⁷¹ ,
What does it measure	Evaluates whether a generated response contains all relevant details necessary to make an accurate conclusion, regardless of how those details are expressed. This focuses on content coverage rather than text similarity.
Considerations for use	- Important for data extraction tasks where missing information can impact review conclusions - Likely requires domain expertise to assess what constitutes "complete" information for the specific evidence synthesis task - Should be distinguished from text similarity metrics as it focuses on information content rather than wording
Example(s) of its use	<i>Sensitivity</i> and <i>completeness</i> of search strategies created using a text-mining word frequency tool (PubReMiner) compared to current best practice in Dullea et al. ⁷²

Factual correctness	
Related terms / metrics	FActScore ⁷³ , Veriscore ⁷⁴
What does it measure	The extent to which outputs generated by an AI system are factually accurate and can be verified against a trusted knowledge base (external source or human verification). This includes detecting hallucinations - content that is fabricated or not based on the input data.
Considerations for use	Factual correctness does not guarantee <i>information completeness</i>

	• Assessment may be complicated by nuance – some claims may be partially true. • Hallucination detection is a key component i.e., should test whether AI generates content that isn't truly present in source material. Can be evaluated by asking about content not in the provided data, or verifying that cited sources exist and contain claimed information • Best applied in combination with complimentary metrics to characterise the severity and potential impact of incorrect outputs (see <i>error severity</i>, <i>information completeness</i>) • Multiple evaluation approaches needed as different types of hallucinations may occur in different contexts
Example(s) of its use	• Testing whether an LLM fabricates data when asked to extract information not present in a study report • Verifying that AI-generated citations actually exist and contain the claimed information

2. Reliability

2.1 Variability

Variation from the mean	
Related terms / metrics	Variance, Dispersion
What does it measure	The extent to which the outputs of an AI system deviate from the mean output (e.g. across multiple tasks or datasets)
Considerations for use	• Strong average accuracy metrics (e.g. 98% recall) can mask much poorer performance on individual tasks or datasets • High variability may suggest inconsistent or unreliable performance
Example(s) of its use	Several evaluations of AI tools to automate citation screening have observed significant variations in performance when applied across different datasets/domains. ^{h 75}

2.2 Consistency

Inter-rater reliability	
Related terms / metrics	Cohen's Kappa; Fleiss' kappa; Intraclass Correlation Coefficient.
What does it measure	<i>Inter-rater reliability</i> measures the degree of agreement among different raters.
Considerations for use	Several factors can influence inter-rater reliability such as: • The clarity of the rating criteria • The experience and expertise of the raters • The complexity of the task Recommended to do pilot testing on a small sample initially to identify the causes of any systematic inconsistencies. See metric overview in Rainio 2024⁵⁶

^h <https://www.covidence.org/blog/ai-screening-automation-systematic-reviews/>

Example(s) of its use	Evaluating inter-rater reliability of LLM screening versus human screeners in Spillias et al. ⁷⁶
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Response stability	
Related terms / metrics	Test-retest reliability
What does it measure	The extent to which outputs of an AI system change when the input remains the same e.g., whether an LLM output changes when submitting an identical prompt multiple times.
Considerations for use	- Generative models often have a “temperature” or variability parameter which can be user-specified to adjust model creativity - Exact measurement comparison parameters will depend on output type (e.g., whether this is text-based, numerical, or binary)
Example(s) of its use	Comparison of human-extracted descriptions of PICOs used in randomised trials with those extracted by an AI tool by Schmidt et al ¹⁹

2.3 Robustness

Input robustness	
Related terms / metrics	Prompt stability
What does it measure	The extent to which an AI system generates comparable outputs when there are superficial differences in the phrasing or structure of input prompts.
Considerations for use	Prompt variations should retain the same meaning as the original prompt – but can differ in exact word choice, order, or tense
Example(s) of its use	- Evaluation of the impact of changing distinct prompts (e.g. reordering, deletions, modifications) on pre-trained language models ⁷⁷ - In an evidence synthesis context, this could involve experimentation to assess the impact of modifying the prompts in different ways for LLM-based data extraction tasks

3. Efficiency

3.1 Resource utilisation

Cost-effectiveness, workload reduction or time efficiency	
Related terms / metrics	Burden
What does it measure	Quantifying the amount of manual work or cost saved
Considerations for use	Requires accurate data on the manual work or cost of specific tasks to be captured before evaluating the AI system. For cost-effectiveness analysis, careful collection of time-on-task data is needed.

	Total workload assessment should consider both time saved during use and time invested in training/setup, including time spent -screening articles to train classifiers -developing and testing prompts -on initial system configuration and learning Stopping criteria for computer assisted screening: With machine learning models that help evidence synthesists prioritise screening (e.g., listing most relevant records first), a stopping criterion is needed to understand when it is safe to stop screening those listed as least relevant (reference) to inform the workload reduction estimate.
Example(s) of its use	• Rapid review and map which estimates reduction of screening burden by different automated tools in Adam et al. ¹⁰ • Cost-effectiveness of a machine learning methods for automated study identification in Shemilt et al. ⁷⁸

3.2 Workload impact

Time-on-task	
Related terms / metrics	Task completion time
What does it measure	The length of time it takes a human to perform a given task.
Considerations for use	It is difficult to collect time-on-task data automatically (e.g., from web use logs) because it is not possible to tell whether an evidence synthesist is actively working on a given record, or has actually switched away to do something else.
Example(s) of its use	• A SWAR protocol for evaluating time-on-task to screen a sample of 100 records using two different approaches by Devane et al. ⁷⁹

Human effort required for verification	
Related terms / metrics	Manual review time
What does it measure	The length of time it takes a human to verify the performance of the AI system.
Considerations for use	• This may be performed on either a subset or the entirety of the classified / generated outputs • Particularly important metric for tasks that will require ongoing “human in the loop” verification
Example(s) of its use	• Used as efficiency metric for different data extraction approaches in Li et al. (77) • Used as an efficiency metric for LLM- based risk of bias assessment and data extraction in Lai et al. ^{45,80}

4. Usability

4.1 User experience

User satisfaction and usability

What does it measure	User satisfaction measures how pleased users are with a tool, considering issues such as reliability, ease of use, and whether it met their expectations. Useability metrics assess how effectively and efficiently users can interact with an AI tool. These metrics also include learnability and accessibility and can include a wide range of data collection techniques including measuring <i>time-on-task</i> (see below), examining error rates, and conducting ‘think aloud’ studies where users verbalise what they are thinking when using a tool.
Considerations for use	Key for user studies is ensuring that the recruitment of study participants is successful. Studying the views of ‘early adopters’ may not provide the developer team with the kind of feedback that they need to evaluate wider satisfaction and utility. See Castro and colleagues⁸¹ who mapped automated tools for usability evaluations.
Example(s) of its use	Cleo et al. assessed usability in an evaluation of four screening tools⁸²

4.2 User integration

Implementation barriers	
What does it measure	Barriers to uptake for an AI system into the systematic review process. These qualitative data may be collected by reviewers as part of a SWAR, using user testing surveys and/or interactive focus group sessions.
Considerations for use	- May be helpful to categorise barriers into key themes (e.g., trust, ethical concerns, technical expertise) - May overlap with <i>interoperability</i> measures if existing systems in place (e.g., at an organisation level) are restrictive
Example(s) of its use	

Interoperability measures	
What does it measure	The extent to which an AI system is (i) compatible with standard file formats and tools commonly used for evidence synthesis tasks (e.g., RIS, Endnote XML, CSV), and/or (ii) compatible with other AI systems (via APIs or other data exchange standard formats).
Considerations for use	- The tools considered to be “commonly used” may differ between research domains e.g., a systematic review platform specialised in laboratory animal reviews versus a platform for clinical reviews - Database export fields and identifiers (e.g., PMIDs) will also differ across research fields

	To evaluate integration with other AI systems, consider whether it is fully programmatic and accessible or only available via a “point-and-click” interface.
Example(s) of its use	Scientific report detailing the implementation of AI systems for systematic reviews at the European Food Safety Authority – including considerations of interoperability within their current processes ⁸³

5. Risk assessment

5.1 Error impact

Error severity scoring	
What does it measure	The potential downstream impact of an error made by an AI system (if left uncorrected).
Considerations for use	- Highly dependent on task context - Less well-studied in review stages beyond data extraction - According to Gartlehner et al ⁸⁴, mistakes during data extraction can be characterised into major errors (could lead to erroneous conclusions), minor errors (influence on data quality, but less severe), and inconsequential errors (no meaningful impact on interpretation).
Example(s) of its use	Gartlehner et al. evaluates LLMs for data extraction and classifies errors into “major” and “minor” errors, hallucinations, and missing data ⁸⁴

Missed critical information rates	
What does it measure	The degree of critical information (e.g., relevant studies not identified and/or included) missed due to use of an AI system.
Considerations for use	This differs from sensitivity as it tries to capture the importance of the information too, but overlaps with 'error severity scoring' above.
Example(s) of its use	Study by Gartlehner et al. ⁸⁵ which measures percentage of missed relevant studies (critical information) when employing single vs dual abstract screening

Changes in evidence synthesis result	
What does it measure	These metrics aim to capture the extent to which the use of an AI tool has resulted in changes to the findings of the evidence synthesis.
Considerations for use	The field is still rather undeveloped, and there are no widely used metrics that encapsulate this concept, but the following example is illustrative of the approaches that might be taken.
Example(s) of its use	Marshall et al. ⁸⁶ re-ran meta-analyses from 2,512 systematic reviews using rapid review methods and evaluated the extent that

	results differed from the published versions in the following ways: changes of pooled odds ratios, changes in statistical significance, changes in effect direction.
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5.2 Safety & ethics

Indicators of bias	
What does it measure	These metrics aim to capture the extent to which the use of an AI tool has resulted in bias
Considerations for use	The field is still rather undeveloped, and there are no widely used metrics that encapsulate the need to evaluate bias. However, Marshall and colleagues assessed this through changes in the assessments of the risk of bias of individual studies.
Example(s) of its use	Marshall et al.⁸⁶ re-ran meta-analyses from 2,512 systematic reviews using rapid review methods and evaluated the extent that results differed from the published versions. They also assessed any changes in the risk of bias assessments.

Transparency measures	
Related terms / metrics	Technical transparency, model transparency, explainability
What does it measure	The extent to which the behaviour of an AI tool can be understood and explained. This may include technical transparency (documentation of training data, model version, model parameters) and explainability (how and why the system provided a certain response)
Considerations for use	- For generative models, this may involve methods to derive a justification for the outputs of an AI system – for example, asking for the text snippet or reference material used to generate an output - Locally hosted, open-source models typically offer more transparency and auditability than proprietary AI systems - Qualitative frameworks and assessment tools can be used to evaluate transparency systematically
Example(s) of its use	Use of textual justification from LLMs employed for structured data extraction from surgical pathology reports⁸⁷

Environmental and sustainability impacts	
Related terms / metrics	Carbon footprint assessment, sustainability metrics, Green AI measures
What does it measure	The environmental costs associated with using an AI tool, including energy consumption, water usage, and carbon emissions related to computational requirements.
Considerations for use	Can be assessed using qualitative frameworks (e.g., AI Ethics frameworks, sustainability assessment checklists) or

	quantitative measures (e.g., energy consumption per task, carbon footprint calculations) • Particularly relevant for LLMs and energy intensive AI systems • Existing frameworks include the GREENER principles for environmentally sustainable computational science⁸⁸, responsible AI frameworks, and institutional sustainability assessment tools • The Green Algorithms project (https://www.green-algorithms.org/) groups calculators and resources that researchers can use to estimate and reduce the carbon footprint of their project. • Should consider both training costs (one-time environmental impact) and inference costs (per use environmental impact) • Important for organisational sustainability commitments and comprehensive ethical AI evaluation • May involve both technical measurements and policy/framework-based assessments
Example(s) of its use	• Comparing the carbon footprint of manual versus AI-assisted systematic review processes • Evaluating different AI models for the same task using sustainability frameworks

Topic-specific safety	
What does it measure	Safety concerns that are specific to a given domain or intended, defined by subject-matter experts who understand how errors in that context could raise safety concerns.
Considerations for use	• Consider what counts as “harm” in this context, who could be affected (e.g. patients, researchers, policymakers), and the severity of different types of errors.
Example(s) of its use	In a comparison of AI assisted and human generated plain language summaries of Cochrane reviews, Devane et al assessed safety concerns specific to public-facing health summaries. ⁸⁹ This included risks of misinterpretation bias/inappropriate recommendations, presentation of uncertainty, and consistency with the source review.